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Evolving Automatic Speech Recognition (ASR) System on The Website for Learning to Facilitate English Speaking Practice

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Abtract

This study explores a product system for speech recognition, which is termed Automatic Speech Recognition (ASR), developed through research and development (R&D) method by adapting the ADDIE model (analysis, design, development, implementation, and evaluation). The product yielded through this research is an ASR system embedded into a website for the practice of speaking English as a foreign language (EFL) for students at the intermediate English level. Before the product was developed, researchers had conducted a need analysis to ensure that the product was needed as a learning support tool for English-speaking practice. After that, the feasibility of the product was measured through the evaluation of two independent experts from two fields of expertise: the expert of information and communication technology (ICT), and the expert of technology-enhanced language learning (TELL). Twenty-two EFL students were invited purposively to assess the ASR system during the trial to users. Results show that the ASR system proves to be a valuable educational tool, offering accessibility, usability, and interactive feedback that enhance user engagement and support language learning. However, this study has a significant problem, which is the inconsistency of speech recognition accuracy, particularly the speech from non-native speakers of English.

Keywords: ASR, speaking skills, WBLL, speech recognition, ADDIE model

INTRODUCTION

The rapid development of information and communication technology (ICT) has significantly influenced educational practices where educational institutions increasingly adopt ICT as a tool to facilitate student learning. In the context of language education, the integration of technology has been introduced since the concept of computer-assisted language learning (CALL) became familiar to language learners. As the earliest approach that leverages computer technology for educational purposes, CALL provides numerous opportunities for foreign language learners to engage in autonomous learning.

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The prominent feature of CALL has provided its accessibility and popularity, particularly in secondary and higher education levels [1]. This development has led to the emergence of web-based learning (WBL), which is also referred to as website-based learning. Specifically, in the domain of language education, it is termed as web-based language learning (WBLL). WBLL has gained widespread recognition, allowing language instructors to extend their teaching practices beyond the traditional classroom setting [2].

Several studies have demonstrated that WBLL can effectively improve speaking skills and even alleviate learners' anxiety when speaking English as a foreign language [3]--[7]. However, while WBLL platforms can provide valuable opportunities for speaking practice, they are not sufficient on their own to ensure learners' mastery of speaking skills. There are still additional critical factors that contribute to effective speaking, such as pronunciation—the ability to correctly produce sounds and words in the foreign language. For Indonesian learners of English, a language that is structurally and phonologically distinct from Indonesian, pronunciation often presents particular challenges. Mispronunciation can hinder communication and fluency. To address this, WBLL platforms can incorporate features to help learners identify and correct pronunciation problems. One technology that is commonly known to recognize anyone's speech is Automatic Speech Recognition (ASR), which converts spoken language into text, allowing learners to receive feedback on coping words to be pronounced.

ASR technology is already integrated into several common commercial language learning platforms, such as *Duolingo, ELSA, Rosetta Stone,* and *TalkieBuddy*. These platforms guide learners to read and pronounce English words or phrases. When learners pronounce the words, the ASR system processes the speech and compares it to the platform's database, which offers feedback on the speech recognition. Several studies have explored the development of ASR application systems in language learning. For instance, an ASR feature was developed for an English for Specific Purposes (ESP) website to evaluate conversation accuracy in job interview scenarios [8]. In addition, a website utilizing ASR technology was developed to assess the English-speaking skills of students and prospective English teachers [9]. Further research focused on the development of an ASR-based language learning website aimed at teaching Malay to non-Malay-speaking children [10]. All of those studies' findings suggests that ASR systems can be effectively used for advanced English-speaking assessments; ASR-powered website can enhance English speaking proficiency; and ASR system can be effectively employed in language learning, even in contexts with limited data.

The aforementioned studies reported the significant potential of ASR in assisting foreign language learners to enhance their speaking skills. By providing immediate feedback on pronunciation and speech accuracy, the ASR system enables learners to self-monitor and adjust their spoken language. These systems are particularly valuable in promoting independent practice, helping learners gain confidence, and improving their speaking abilities in a supportive and low-stress environment. Therefore, this study evolved the ASR system in a previously developed website for English speaking practice. To explore its potential, this study aims at (1) examining the need for an ASR system for speaking practice; (2) developing the ASR system into the website by the design for English speaking practice; (3) evaluating the implementation of the ASR from experts' assessment and users' evaluation.

Literature Review

Web-based Learning (WBL)

The WBL is defined as "education that takes place solely through the web, without the use of physical learning materials or direct face-to-face contact" [11]. This description emphasizes that in pure online learning, the web is the primary medium for both content delivery and student interaction. On WBL platforms, students can engage in learning independently or collaboratively, regardless of time and location, making the learning process more flexible and accessible.

The web platform offers flexibility in teaching and learning by removing the physical limitations of the classroom and the time constraints imposed by traditional schedules [12]. Additionally, web-based multimedia learning experiences enable lectures and skill

demonstrations to be shared with students remotely. As a result, college and university learning resources can be enhanced and more widely distributed via the world wide web (www), which leads to a shift in focus from teaching to learning and from instructors to students.

From a learner's perspective, one of the main advantages of WBL is that it provides students with access to authentic English-language resources. As noted, the internet predominantly operates in English, making it a highly motivating factor for learners who are eager to improve their proficiency in the language [13]. Many online platforms offer a variety of tools for practicing grammar, pronunciation, vocabulary, speaking, and listening skills. In the context of language learning, WBL platforms are often referred to as WBLL systems [14]. These platforms specifically accommodate language learners, offering targeted resources and interactive features to enhance their linguistic competence.

Web-based Language Learning (WBLL)

WBLL has become a significant and widely discussed concept in the field of language education [15]. WBLL offers learners the opportunity to engage with challenging and complex communication tasks, providing them with a sense of autonomy in their learning process [16]. This aligns with an assertion that technology-based learning activities play a pivotal role in shaping learning outcomes [17].

Numerous studies have demonstrated that WBLL platforms are particularly effective for enhancing speaking skills. For instance, a study investigated the impact of web-based artificial intelligence (AI) on learners' anxiety and proficiency in English speaking [3]. The study utilized an AI chatbot in interactive tasks, which facilitated increased oral communication and interaction. The findings suggest that integrating AI technology into web-based platforms offers a flexible, interactive, and student-centered approach to learning. Similarly, the other research explored the use of a web-based application called VoiceThread, an asynchronous audio-conferencing tool, and found that it helped learners of English have a more relaxed and enjoyable learning environment, which, in turn, boosted motivation and decreased the anxiety typically associated with speaking [4].

Further research examined the development of a website designed to assist vocational school students in learning English pronunciation and vocabulary [7]. The study confirmed that WBLL not only enhanced speaking skills but also contributed to lower anxiety levels when speaking in English. Collectively, these studies demonstrate the significant benefits of web-based technology as a tool for language learning, particularly in fostering speaking proficiency. Technological tools, as observed such as web-based platforms, can create an engaging and supportive learning environment [5]. These platforms help overcome a major barrier to language acquisition, thus facilitating more effective English language learning.

Automatic Speech Recognition (ASR) Technology

ASR technology has had a profound impact on education, offering valuable tools that enhance both teaching and learning. In the context of language acquisition, ASR plays a crucial role in helping learners improve pronunciation and practice speaking skills in various contexts [18] [8]. One notable example of ASR's application in education is the development of the Voice Geometry Painter, a tool that enhances learning by making it both easier and more enjoyable [19]. A variety of ASR tools are available, including *Dragon Dictate, Dragon Go, Google Now, SIRI, MeMeMeMobile, SILVIA,* and *Vlingo*, each with unique features designed to support educational purposes [20]. Two of the most widely recognized ASR engines for language learning are *Apple's Siri* and *Google Speech Recognition (GSR)*, both of which can be integrated into Learning Management Systems such as Moodle [21].

ASR technology is a key component of CALL systems, which utilize advanced computing technologies for natural language processing (NLP). These ASR-based CALL systems have gained significant attention from researchers and educators, particularly in the field of English language teaching, due to their ability to evaluate learners' speech in real time, which is highly beneficial for improving listening and speaking skills [22] [23]. For example, a study utilized audio-visual materials integrated into an ASR system to diagnose pronunciation errors [24], while another

study incorporated ASR technology into e-learning software to enhance speaking and listening skills in college classrooms [25]. Both studies demonstrate that ASR technology is highly effective in language learning, allowing learners to practice speaking without the stress or pressure often associated with traditional classroom settings.

The findings from these studies highlight ASR technology offers significant benefits in language education, particularly in supporting speaking assessments (for teachers) and enabling learners to evaluate their pronunciation errors and improving speaking fluency in a more relaxed environment. Overall, ASR technology proves to be an invaluable tool in education, facilitating language learning and providing crucial support for learners in areas such as pronunciation, vocabulary, and speaking skills.

METHOD

Research Design

The research and development (R&D) method was used in this study to create a specific product and to assess its effectiveness [26]. The product developed was a technological tool in the form of ASR, which was embedded in a previously developed website named "Harati Speaking" for a speaking practice platform [27]. This research employed the ADDIE model, as outlined by Dick and Carey [26], in the context of advancing the ASR system through the R&D method. The ADDIE model consists of five stages: Analysis, Design, Development, Implementation, and Evaluation. Through this model, researchers examined the need analysis, developed the ASR design, and assessed expert evaluations and user experiences related to the ASR product.

Participants

There were 22 students of second year study selected to participate in this research taken from piloted classroom of intermediate English-speaking course in the English Language Education Program of the University of Palangka Raya. Those respondents contributed to the data of need analysis and users' experiences related to the ASR system. Furthermore, the other participants of this research were two experts, respectively one from the field of ICT and one from technology-based English learning. Those two experts contributed to the data of product evaluations on the ASR system used in the website for speaking practice.

Instruments

There were three phases of data collection to support this research. In the first phase, researchers collected the data from students for a need analysis to inform the design of the ASR product in supporting English-speaking practice. The instrument used for collecting the data was a 10-item questionnaire to rate whether the ASR system was required on the website for speaking practice. Two indicators were contained in the questionnaire: the relevance and the effectiveness of the ASR system for speaking practice.

In the second phase, data were collected from experts to validate the feasibility of the ASR product before its implementation to students as users. The instrument used in this phase was a Likert-scale questionnaire with responses ranging from 1 to 4. Two domains of questions in the questionnaire with 10 questions for each specialized expert. The ICT expert evaluated the ASR feature's capacity to support learning, while the language learning expert assessed its effectiveness in enhancing speaking practice.

The last phase of data collection is the evaluation of user experiences from product implementation. The instrument used for obtaining the data was a user satisfaction questionnaire adapted from Lund [28], a widely recognized tool for assessing interactions with technology-based educational products. Three dimensions of product were evaluated: ASR usability, user satisfaction, and learning outcome support, with five questions for each, which means 15 questions in total.

Data Analysis Procedures

This study employed a combination of descriptive quantitative for calculating data by percentages. For the data of "needs analysis", the questionnaire responses were analyzed to calculate the percentage of positive and negative responses that provided users' needs of ASR systems in the website for English speaking practice. This analysis focused on identifying the tendencies and preferences of users related to the relevance and effectiveness of the ASR system. Then, data from "expert validation" were analyzed using descriptive techniques by calculating ratings provided by two experts and comparing both data to assess the feasibility of the ASR product. This evaluation considered the dual aspects of ICT and technology-based language learning product assessments for English learning. Last, data from "user experiences" were further analyzed to calculate the mean score for the three assessed dimensions of the product—ASR usability, user satisfaction, and learning outcome support. The findings were then interpreted concerning whether ASR is useful for speaking practice, whether ASR is satisfied users, or ASR supports learning outcome, particularly English-speaking skills.

RESULT AND DISCUSSION

The research and development (R&D) study focused on the development of an Automatic Speech Recognition (ASR) feature integrated into a web-based English learning platform aimed at enhancing the speaking skills of students in the English Language Education Study Program. The results are summarized into three parts as indicated in the objective: (1) the need analysis of ASR system for speaking practice; (2) developing the ASR system design for English speaking practice; and (3) evaluating the implementation of the ASR from experts' assessment and users' evaluation.

The need analysis of the ASR system for speaking practice

Need analysis is an analysis to assess the necessity of the ASR system embedded in a learning website for improving speaking skills. This stage identified key requirements for the effective integration of ASR in the learning process within three indicators: relevance, effectiveness, and benefits of the ASR system for speaking practice. The following are the data of need analysis derived from questionnaires administered to students as the users in the course of Intermediate English Speaking.

Table 1. Results of needs analysis on the relevance of ASR system

No.	Questions	Number of answers	
		Yes	No
1.	Do you think using the ASR system will help you improve your English-speaking skills?	21	1
2.	Does the ASR system meet your needs to practice speaking English?	19	3
3.	Would it help if the ASR system provided guidance on pronunciation in English?	22	0
4.	Do you feel that ASR can make learning English speaking more engaging than traditional methods?	22	0
5.	Are you interested in using technology-based tools to help you practice speaking English?	22	0
	Average percentage (n=22)	96%	4%

The needs analysis, as shown in Table 1, revealed that a significant majority of students (95%) believed that ASR could improve their speaking skills, with 86% considering it highly relevant to their practical needs. Moreover, all participants (100%) agreed that the pronunciation guidance provided by the ASR system would be beneficial. They unanimously expressed that incorporating the ASR system would make learning English more engaging than traditional

methods. Additionally, the average percentage showed that 96% of responses were positive. This finding suggests that the ASR system holds strong relevance and potential in enhancing students' speaking proficiency through speech recognition feedback.

Table 2. Results of need analysis on the effectiveness of ASR system

No.	Questions	Number of answers	
		Yes	No
6.	Do you think the ASR system can help you recognize your mispronunciation?	19	3
7.	Will the feedback from the ASR system encourage you to improve your English-speaking skills?	20	2
8.	Do you feel that the ASR system will help you understand your current speaking skill level?	20	2
9.	Is it useful if the ASR system provides immediate feedback on your English pronunciation?	22	0
10.	Are you confident that the feedback provided by ASR will be accurate and helpful to improve your speaking skills?	18	4
	Average percentage (n=22)	90%	10%

Table 2 indicates positive perceptions among students about the effectiveness of the ASR system. Most participants (86%) believed that the ASR technology would help them identify mispronunciations, while 91% felt that the feedback from ASR would motivate them to refine their speaking abilities and to understand their current proficiency levels. All participants (100%) acknowledged the advantages of receiving direct pronunciation feedback for their skill development. Furthermore, 82% expressed confidence in the feedback's accuracy and its ability to enhance their speaking skills. The average percentage responses showed a 90% affirmation rate, which highlights the ASR's effectiveness in supporting pronunciation improvement and encouraging greater confidence from the feedback received. In summary, the result of the need analysis highlights a strong interest and confidence among students in the relevance and effectiveness of ASR technology as a tool to support English-speaking practice.

Developing ASR product design

Since the responses of the need analysis are highly positive, researchers developed the ASR design for speaking practice by providing speech recognition to support the website. The ASR system in this research is a feature developed by Mozilla developer, which can be obtained at https://developer.mozilla.org. The platform provides a code for proposing the ASR system, and from the code given, it can be embedded into the researchers' website that has been developed. In this research, the ASR system can be used to detect mispronounced words in English speech. Thus, the instructors or teachers can easily and accurately provide speech corrections as feedback to their students. The following is the flow system of the ASR system design that was developed in this research.

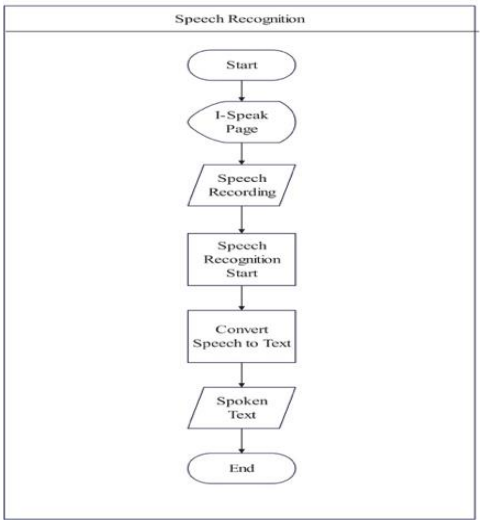


Figure 1. System flow of ASR system

As illustrated in Figure 1, the speaking practice process begins when the learner accesses the website and enters the "I-Speak" menu on the web platform. Upon initiating the practice session, the learner's speech is recorded in the "Speech Recording" module while simultaneously being processed by the ASR system in the "Speech Recognition Start" stage. Once the recorded speech is transmitted to the database, the ASR system converts it into text within the "Convert Speech to Text" module. At this stage, the transcribed text is displayed on the screen as "Spoken Text," which provides the learner with a visual representation of their speech output.

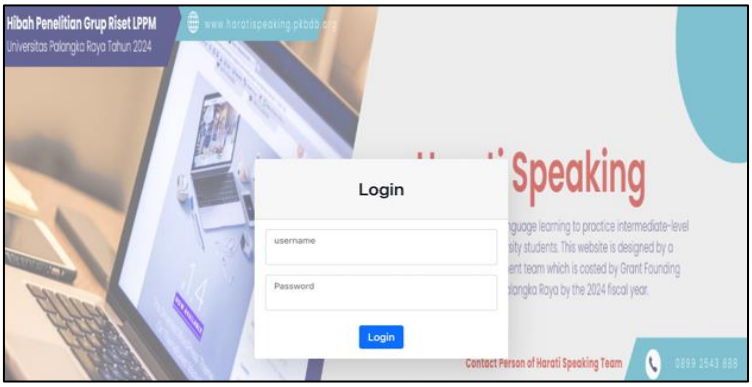


Figure 2. The login frontpage of the website

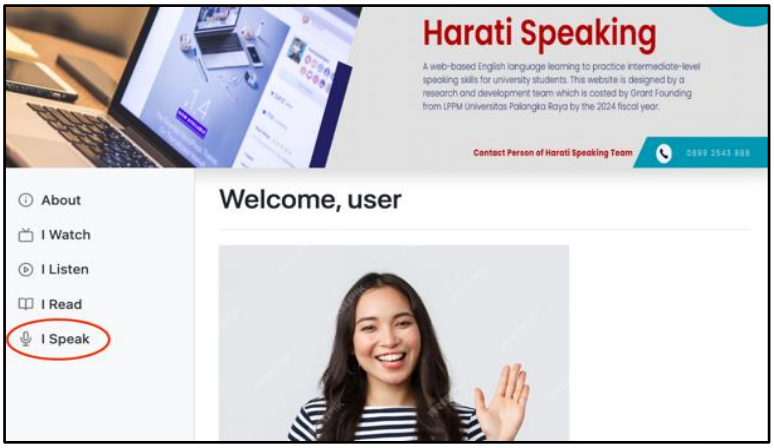


Figure 3. The access to run ASR system

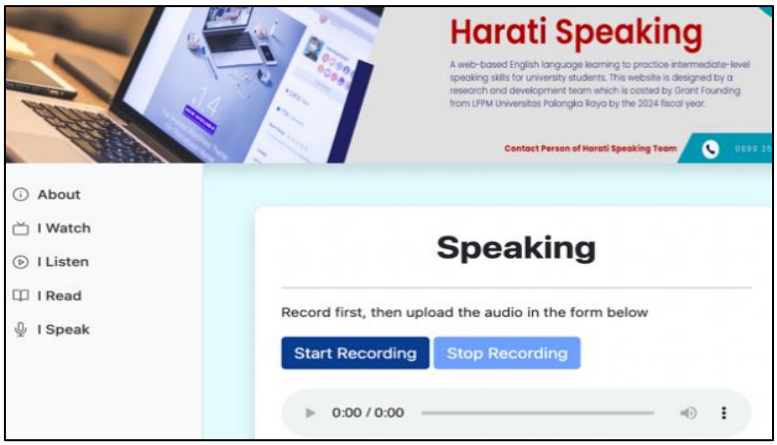


Figure 4. The audio recording display interface

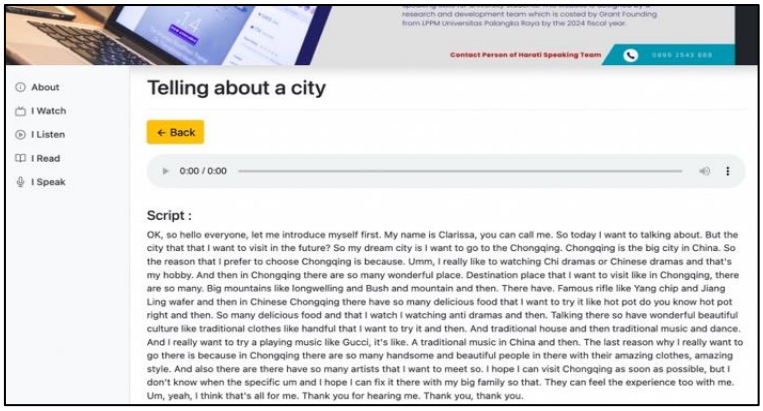


Figure 5. The speech-to-text results by the ASR system

In the design developed from the system flow diagram (Figure 1), users must first go to the website to log in (Figure 2), and they come into the bar menu of "I Speak" (Figure 3). Then, the ASR system generates two types of data, first, the learner's speech is recorded into the audio recording (Figure 4); and second, the speech input in the audio is recognized and transcribed into text (Figure 5). Both types of data serve as valuable resources for learning evaluation that enable lecturers to assess and improve the learner's pronunciation and speaking proficiency effectively.

The evaluation of ASR system implementation

There were two types of evaluation on ASR system implementation: first, the product was validated by two related experts, and then it was rated by users—students who took the course of intermediate English speaking. The validation from experts was conducted after the product design was developed, while the evaluation from users was done when the product was tested to the students as users of the product.

Expert Validation

Data in Table 4 below provides a thorough evaluation of the ASR system on the website, which aims to support the practice of speaking English.

Table 4. Validation Results from the ICT expert

No.	Validated indicators	Scores
1.	The ASR system on the website is reliable and accurately converts speech into text.	3
2.	The speed at which the ASR system processes user input is sufficient for real-time feedback during speaking practice.	3

3.	The ASR system shows compatibility with multiple browsers and devices, and improves accessibility.	3
4.	The ASR system uses powerful algorithms that support effective speech recognition for English learners.	4
5.	The ASR feature on this website is secure and protects user data, especially voice recordings.	2
6.	The ASR system integrates seamlessly with other web features providing a seamless learning experience.	2
7.	ASR systems allow for updates and upgrades without causing any disruption to the user experience.	4
8.	ASR features include advanced functionality (e.g., voice modulation analysis) that is useful for learning English pronunciation.	4
9.	ASR systems can be adapted to different proficiency levels (e.g., beginner to advanced).	3
10.	The ASR system has low latency, ensuring minimal delay between speaking and feedback, essential for effective practice.	2
	Average score achieved	3.0

From the validation results, a score of 3 was given for the ASR system's reliability and effectiveness in converting speech into text. It is similar to providing real-time feedback where ASR also gets a score of 3, in which the system can give feedback to users by indicating adequate processing speed. The system's compatibility with multiple browsers and devices is given a score of 3, which highlights its accessibility across platforms. The expert also noted the strength of the algorithm used, which gave it a score of 4. This validation score is worth it to its performance as it effectively supports English speech recognition, thus benefiting users. However, the ASR system receives a lower score of 2 for data security, which indicates some concerns regarding the protection of user voice recordings. Integration with other website functions for a seamless learning experience also has a score of 2, which indicates minor validation in this indicator. This ASR system runs well, with a score of 4 in its capacity to handle updates and upgrades without disrupting the user experience. Advanced functionality, such as voice modulation analysis, receives a high score of 4 for this feature. The ability to adjust to suit different skill levels, from beginner to advanced, has a score of 3, indicating that it can be adapted to the diverse needs of users. However, this ASR system receives a lower score of 2 in terms of latency, which indicates there may be a delay in the feedback given during the product test session.

As a result, the ASR system embedded on the website generally performs well, with strength in durability, advanced functionality, and multi-platform compatibility, with an average score of 3.0. However, there are still weaknesses shown in terms of improved data security, seamless integration with the website, and latency, which can affect the user experience during speaking exercises.

Table 5. Validation results from the expert of technology-based English learning

No.	Validated indicators	Scores
1.	The ASR system provides accurate feedback on mispronunciation in English that helps learners correct themselves.	3
2.	The ASR feature allows users to practice English pronunciation repeatedly which improves retention and skill development.	4
3.	ASR system feedback is understandable and actionable in guiding learners on how to improve their speaking skills.	2

4.	The ASR feature on the website motivates learners and encourages regular English-speaking practice.	3
5.	The ASR system recognizes and adapts to different English accents, and supports a wide range of users.	3
6.	The ASR system identifies and distinguishes minor pronunciation errors that English learners can improve.	3
7.	The ASR feature helps learners build confidence in their spoken English by providing constructive feedback.	4
8.	The ASR system supports interactive exercises that increase learners' fluency in spoken English.	3
9.	ASR feature aligns with language learning goals to improve pronunciation and fluency.	2
10.	The ASR-based feedback provided on the website is consistent with feedback from human instructors, so it can improve its reliability.	3
	Average score achieved	3.0

Validation from the expert of technology-based English learning, as shown in Table 5, the ASR system is scored 3 in its ability to provide accurate feedback on pronunciation errors to assist learners in self-correction of their speech. In addition, the ASR allows for repetitive English pronunciation exercises, so it is very beneficial for skill development and retention where this indicator has a score of 4. However, ASR's feedback in guiding users on effective ways to improve speaking skills has received a lower score of 2 since there is no instruction or suggestion from the system. Moreover, the ability of the ASR system to motivate learners and promote regular English practice has a better result with the score of 3. The ASR can recognize and adapt to a variety of English accents, which supports different users' mother tongue, with a score of 3, too.

Meanwhile, this ASR system can identify small pronunciation errors that can be corrected by students themselves, with a score of 3. To help build learners' confidence, the ASR system provides constructive feedback with a score of 4 for this supporting role. Furthermore, ASR supports interactive exercises that encourage speaking fluency with a score of 3. However, the alignment of the ASR system with overall language learning goals, especially in improving pronunciation and fluency, only obtained a score of 2. This suggests that some adjustments may be needed to meet the learning objectives. Finally, the consistency of the ASR system's feedback with human instructors is positively noted, with a score of 3 for its reliability in mirroring feedback as the human instructor does.

Thus, the ASR system on this English learning platform performs well in encouraging skill development and supporting interactive practices, with an average score of 3.0. However, improvements can be made in the clarity of feedback, alignment with language learning objectives, and actionable guidance to increase their effectiveness in helping students improve their English pronunciation and fluency.

User experience

The ASR product has been tested three times in three weeks of product trial to the students who took the course of intermediate English speaking. After the third trial was completed, users were given a questionnaire to ask their experiences while using the ASR system for practice speaking during the three weeks. The questionnaire has evaluated three aspects of the ASR system, namely the usefulness of ASR (usability), satisfaction with ASR (user satisfaction), and learning outcomes using ASR (learning outcome). The questionnaire has a score range of 1 to 5, where scores ranging from 1.0 to 2.9 indicate a negative experience, scores ranging from 3.0 to 3.9 indicate a neutral experience, and scores ranging from 4.0 to 5.0 indicate a positive experience.

(1) ASR's usability

The following are the results of user evaluations of the usefulness indicators of ASR product on the website when used during the trial.

Table 6. User Experience Based on Product Usability

No.	Usability indicators	Average score (N=22)	Category
1.	The ASR system on the website is easily accessible.	4.2	Positive
2.	The ASR system accurately recognizes my speech.	3.4	Neutral
3.	I didn't experience any technical issues when using the ASR system.	2.2	Negative
4.	The ASR interface for me to use is very intuitive.	3.4	Neutral
5.	The ASR system gave feedback quickly after I finished speaking.	3.7	Neutral

Based on the findings from the usability of ASR, as shown in Table 6, the accessibility of the ASR system receives the highest score of 4.2, categorized as positive. This indicates that users find this speech recognition technology easily accessible, which reflects a well-designed system for the users. On the other hand, the ASR's ability to accurately recognize speech is rated neutral, with an average score of 3.4. This score suggests that while some users are satisfied with the speech recognition capabilities, others experience challenges. This is likely due to the system's limitations in handling various speech patterns or accents. Similarly, the intuitive ASR interface also received a neutral rating of 3.4. This shows that while some users find the interface easy to navigate, others face difficulties, so there is still a need for an easier and user-friendly design.

The system's ability to provide feedback quickly after the user has finished speaking is rated slightly above neutral, with a score of 3.7. This suggests that users generally appreciate the responsiveness of the system, albeit occasional delays may have hindered a seamless experience. In contrast, the lowest-rated usability indicator is the absence of technical problems, which scored 2.2 and is categorized as negative. This indicates that users often experience technical issues, such as crashes or glitches, that significantly affect their overall experience.

In summary, the finding in terms of ASR's usability highlights several strengths. Ease of access and relatively quick feedback are important strengths that contribute positively to the user experience. However, a low score for technical stability is a significant issue that requires immediate attention.

(2) User satisfaction on ASR

The following are the results of user satisfaction indicator to ASR system on during the product trial.

Table 7. User satisfaction with the ASR system

No.	User satisfaction indicators	Average score (N=22)	Category
6.	I am satisfied with the ASR system installed on the website.	4.0	Positive
7.	I am satisfied with the feedback provided by the ASR system to improve my speaking skills.	3.6	Neutral
8.	I found the ASR system interesting and motivating for speaking practice.	4.1	Positive

9.	I would likely recommend the ASR system to others who want to practice speaking English.	4.3	Positive
10.	Comparison of ASR system with other tools I use for speaking practice.	3.7	Neutral

Table 7 provides insight into user satisfaction with the ASR system designed for English-speaking practice. These findings reveal varying levels of satisfaction across different aspects of the speech recognition technology. Users generally express positive feelings about the performance of the ASR system, as evidenced by an average score of 4.0 for overall satisfaction with this technology. This score indicates that most users find the ASR system reliable and aligned with their expectations.

The feedback provided by the system to improve speaking skills received a neutral rating, with an average score of 3.6. This indicates a gap in user perception, while some find the feedback helpful, others may find it insufficient for real improvement. Similarly, the ASR system was rated positively with a score of 4.1 for being engaging and motivating. This reflects its ability to encourage active participation and sustained interest in speaking practice.

The main strength of the ASR system lies in its high recommendation score of 4.3, which is the highest among indicators. This indicates a high likelihood that users will support this technology system to others who want to improve their English skills. The users underline the perceived value and effectiveness of this speech recognition system. On the other hand, comparing the ASR system with other speech tools resulted in a neutral score of 3.7. This suggests mixed opinions, with some users viewing it as equal to or excellent to other existing tools, while other users may find its capabilities less competitive.

To summarize, while the ASR feature shows a lot of potential and has received generally positive feedback, its value can be further increased by concentrating on improving the quality of feedback and competitiveness with other features. Thus, the ASR system developed by this study can be a more effective and widely recommended resource for English-speaking practice.

(3) Learning outcome

The following are the results of user evaluation of the learning outcome indicator on the ASR system.

Table 8. Learning outcome indicators

No.	Learning outcome indicators	Average score (N=22)	Category
11.	Using the ASR system improved my English pronunciation.	4.0	Positive
12.	The ASR system helped me improve my overall fluency in English.	3.6	Neutral
13.	The ASR system boosted my confidence in speaking English.	3.6	Neutral
14.	I feel that the ASR system is an effective way to practice English conversation.	4.1	Positive
15.	I am satisfied with my progress in speaking English after using the ASR feature.	4.0	Positive

Table 8 reveals positive results where an average score of 4.0 for improving users' English pronunciation indicates that users find the ASR system effective in helping them perfect their English pronunciation skills. The ASR system also received a positive score of 4.1 for its effectiveness in practicing English conversation. This suggests that users view this speech recognition technology as a helpful resource for engaging in conversational practice, which is

essential for developing practical speaking skills. In addition, satisfaction with the speaking progress after using the ASR system scored 4.0, which emphasizes the effectiveness of this speech recognition system in supporting the development of learners' speaking skills.

Despite these positive indicators, users also expressed a neutral perception by the score of 3.6 regarding the role of the ASR system in improving overall fluency of English and increasing confidence in speaking English. Both scores indicate that they may not fully address all aspects of language learning, such as improving fluency and building confidence. Some of these things reflect an opportunity to improve the capabilities of the ASR system to make it a more comprehensive language learning tool for speaking practice.

DISCUSSION

Regarding the result of this study, the findings reveal the strengths and limitations of the ASR system as a tool for English-speaking practice. Several previous studies are in line with this study. When it comes to the product usability, users generally find the ASR system accessible and easy to use. This reflects the importance of intuitive and user-friendly design in educational technology to improve user engagement and satisfaction [29]. However, the inconsistency in speech recognition accuracy indicates persistent difficulties in addressing various speech patterns and accents. This problem has become a concern noted in previous studies where non-native pronunciation frequently impedes ASR performance [30]. Additionally, frequent technical issues, such as system crashes and glitches, underscore the critical need for system stability, as technical difficulties can disrupt the learning process and reduce user satisfaction [31]. Different perceptions regarding the intuitive interface and feedback response further emphasize the need for continuous improvements in responsiveness and system design to better meet user expectations and provide a seamless learning experience [29].

Regarding data of user satisfaction with the ASR-based website in English-speaking practice, users' responses highlight ASR effectiveness and areas for improvement. Several findings from previous research generally find that ASR was a reliable and effective system to support language learning. For instance, a study related to ASR systems improved users' accessibility and usability by providing direct and interactive feedback while the tool was used for speaking practice [32]. In addition, the system's capacity to engage and enhance users has reinforced another study's findings that emphasized the role of ASR in improving learner engagement through the interactive tool that encourages consistent practice [33].

However, some limitations persist, particularly concerning the quality of feedback and its comparison with other tools. Feedback is an important aspect of language learning. As highlighted by the other study [34], users frequently require exact and implementable corrections to facilitate their progress in practice. The system's inability to provide consistent feedback can reduce the effectiveness of the features working in the system. The limitation of ASR in this study is to summarize that ASR systems are indeed valuable, but their performance often varies and does not necessarily outperform traditional learning methods [35].

In relation to the data of learning outcomes, the results underscore the power of the ASR system. The high success rate for better pronunciation and conversational practice demonstrates its fundamental ability to support language learners with targeted language skills. These findings are in line with a study finding that ASR technology effectively supports pronunciation and conversational skills by offering interactive real-time feedback [36]. This evidence validates the effectiveness of ASR-based website systems as a valuable tool for improving certain aspects of language learning, especially for speaking practice.

However, the undeniable score for users' fluency and confidence indicates the limitations of the ASR system designed by this study. Fluency involves not only pronunciation but also the ability to speak fluently and spontaneously, which may require exposure to a wider range of interactive scenarios. Similarly, building confidence is a complex process that often benefits from face-to-face interaction and real-world practice, areas where ASR system evolved for this study may fail. These findings are consistent that while ASR systems are effective in certain language areas, they may not fully replicate the benefits of human interaction [37].

CONCLUSION

The findings of this study highlight both the benefits and limitations of the ASR-based system evolved for English-speaking practice. On the one hand, the system proves to be a valuable educational tool, offering accessibility, usability, and interactive feedback that enhance user engagement and support language learning. It effectively improves pronunciation and conversational skills that strengthen the role of ASR technology in providing speech recognition assistance to English learners. On the other hand, this study also reveals challenges that need to be addressed. The inconsistency in speech recognition accuracy, particularly for non-native speakers, remains a significant problem. Additionally, technical issues such as system faults and crashes hinder the learning experience. The quality and reliability of feedback also present limitations, as ASR-generated corrections may not always be precise or actionable. Furthermore, while the system aids in recognizing mispronunciation of words, it does not fully support the development of fluency and confidence. The system often requires more dynamic and interactive learning features or website interface to enhance users' fluency and to build users' confidence.

Overall, while the ASR system developed in this study offers notable benefits for language learners, continuous improvements are necessary to enhance its accuracy, stability, and feedback mechanisms. Future research and development should focus on refining ASR technology to better accommodate diverse speech patterns, provide more reliable feedback, and integrate features that promote fluency and confidence like in real-world communication. By addressing these challenges, ASR-based learning tools can become even more effective in supporting language acquisition, especially English as a foreign language.

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